Forecast Uncertainty in Learned Metro Dispatch: A Placebo-Controlled Study at a Fixed Training Budget

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Abstract

Supplying a learned dispatch controller with forecast uncertainty may improve service, but the signal's information must be distinguished from the effect of additional inputs.

We reanalyse stored evaluations of maskable proximal policy optimisation in a ridership-grounded metro simulator. Three arms share architecture, input width, reward and a requested 60,000-step training budget; their uncertainty channel contains zero, resampled dispersion or calibrated dispersion. Ten training seeds were evaluated on 30 held-out simulation seeds across four scenarios. The original two-test family uses training-seed means and Holm correction.

Under surge, the uncertainty-aware controller's left-behind fraction was 0.031681, versus 0.017939 for the point-forecast controller (mean difference +0.013742; adjusted p = 0.007741). The paired Hodges–Lehmann estimate was 0.013588 (95% BCa interval [0.006570, 0.018523]). It also performed worse than the placebo arm (difference +0.008525; adjusted p = 0.011707). A tuned threshold heuristic descriptively outperformed every learned arm in all four scenarios.

The original hypothesis was not supported under this configuration. The result does not distinguish ineffective information from inadequate optimisation or training. Missing original model/data artifacts prevent complete experiment reproduction and the planned budget follow-up. Registration chronology is internally documented but not independently verified from the available history. Conclusions are limited to the tested simulator and configuration.

Keywords: metro train induction; reinforcement learning; probabilistic demand forecasting; forecast uncertainty; placebo control; reproducibility; nonlinear service loss; negative result; action masking; simulation-based evaluation

1. Introduction

Metro systems absorb demand that varies sharply within the day and irregularly between days. Train induction — the decision of how many trainsets to release into revenue service in

each control interval — sits directly on the trade-off between passenger service and operating resources. Releasing too few trainsets leaves passengers unable to board; releasing too many raises energy and crew cost without commensurate service benefit. Because demand is known only through a forecast, the decision is made under uncertainty (Yin et al., 2016).

Classical treatments of railway rescheduling formulate the problem as a mathematical program and re-solve it as conditions change (Cacchiani et al., 2014; Yin et al., 2017). Reinforcement learning has been proposed as an alternative for settings where re-optimisation is too slow for the minutes available to a controller (Šemrov et al., 2016; Tang et al., 2025), and learned controllers have been applied to rail transit rescheduling and passenger inflow control (Ying et al., 2024; Jiang et al., 2018).

In parallel, probabilistic demand forecasting has matured to the point where calibrated predictive intervals are routine rather than exceptional (Wang et al., 2024; Liu et al., 2026). The natural conjunction — give the learned controller the uncertainty, not only the point forecast — is intuitive, and uncertainty-aware and risk-sensitive formulations are an active area (Lockwood and Si, 2022; Clements et al., 2019).

That conjunction is difficult to evaluate cleanly. An arm that receives an additional input differs from one that does not in more than the *semantics* of that input: it has a different input distribution, different first-layer parameters and a different optimisation landscape. A measured difference is therefore consistent with three explanations — the uncertainty information itself, the additional dimensionality, or seed-to-seed variation in training, the last of which is well documented in deep reinforcement learning (Henderson et al., 2018; Colas et al., 2018).

Research question. Does incorporating calibrated demand uncertainty into a PPO-based metro dispatch controller improve passenger-service performance under stochastic and capacity-binding demand, relative to a point-forecast controller and to controlled baselines?

Hypothesis (registered). The uncertainty-aware controller reduces the fraction of passengers left behind under a demand-surge scenario, relative to an otherwise identical point-forecast controller.

Design. We answer the question with an internally specified, placebo-controlled experiment in which the uncertainty channel's *content* is the only manipulated variable, the training seed is the unit of analysis, and the analysis plan is fixed before the confirmatory runs.

Contributions.

1. A controlled metro-dispatch evaluation framework coupling probabilistic demand forecasting, an explicit uncertainty representation, a simulation environment and reinforcement learning, with classical and heuristic baselines evaluated under common random numbers.
2. A controlled comparison between point forecasts and informative uncertainty using matched scenarios and a **placebo** uncertainty condition with the same channel width. The placebo resamples past dispersion values; it is not an exact permutation control.
3. A statistically robust empirical finding that the tested uncertainty-aware controller performs **worse** than the point-forecast controller on the registered primary endpoint (+0.013742; Holm-adjusted p = 0.007741), reproduced in all four scenarios, in 9 of 10 training seeds, and in 12 of 12 sensitivity conditions.
4. A behavioural association between uncertainty-channel content, episode dispatch volume and service outcomes, alongside an explicit account of what these summaries cannot identify about timing, causality or optimal actions.
5. An evaluation framework and stored result summaries permitting arithmetic reanalysis, with missing historical artifacts and limits to full reproduction disclosed.

We do not claim the framework is novel in its components; each is standard. What the study

contributes is the control structure that makes the comparison interpretable, and the finding that structure produced.

2. Related work

Metro train scheduling and rescheduling. Real-time railway rescheduling has an established optimisation literature, surveyed by (Cacchiani et al., 2014) and (Cordeau et al., 1998). For metro systems specifically, (Yin et al., 2017) formulate demand-oriented train scheduling as mixed-integer linear programs balancing energy efficiency against waiting time, and (Yin et al., 2016) address rescheduling under *uncertain, time-variant* passenger demand using approximate dynamic programming. The latter is a directly related treatment of demand uncertainty in metro scheduling; it differs from the present work in optimising an explicit model of the uncertainty rather than supplying uncertainty as an input to a learned controller.

Reinforcement learning for transit scheduling and control. (Šemrov et al., 2016) apply Q-learning to train rescheduling on a single-track railway evaluated on a real network. More recent work applies deep and multi-agent reinforcement learning to real-time railway rescheduling (Tang et al., 2025), to rail transit services with short-turnings under disruption (Ying et al., 2024), and to coordinated passenger inflow control in peak hours (Jiang et al., 2018). The common motivation is that exact re-optimisation is too slow for the response times a dispatcher has in some applications. Here we focus on matched uncertainty-channel comparisons and report the observed negative result alongside classical baselines that outperform the learned arms.

Uncertainty-aware and risk-sensitive decision making. Reviews of uncertainty in deep reinforcement learning distinguish aleatoric from epistemic uncertainty and survey estimation methods (Lockwood and Si, 2022); risk-sensitive formulations replace expected return with tail-aware objectives (Clements et al., 2019). Our design differs in an important respect: we do not modify the objective or the algorithm. We hold the learner fixed and compare observation constructions without changing to a risk-sensitive method. The imperfect bootstrap control limits isolation of information quality.

Demand forecasting for transit. Uncertainty quantification for travel demand is well developed, including probabilistic graph neural networks (Wang et al., 2024) and mixture-of-expert models for multi-modal demand (Liu et al., 2026). We use a quantile gradient-boosted forecaster; our forecasting contribution is calibration discipline and the freezing of the uncertainty definition before policy training, not model architecture.

Convex loss under demand uncertainty. That a convex cost makes expected cost under stochastic demand exceed cost evaluated at expected demand is the classical newsvendor observation (Arrow et al., 1951), extended to capacity and inventory networks by (Van Mieghem and Rudi, 2002). We invoke this as established theory rather than deriving it, as motivation for future analysis; the historical binned association does not establish structural convexity in our environment (Section 12).

Simulation and digital twins for transit operations. Simulation-based evaluation of rail operations is widespread, and digital-twin approaches in rail have been reviewed systematically (Zhang et al., 2025). We use the term *simulation environment* rather than digital twin, because only a minority of our environment's quantities could be validated against published operating statistics (Section 5).

Controlled and reproducible machine-learning experimentation. (Henderson et al., 2018) document that random seeds alone can produce statistically distinguishable performance from identical code, and (Colas et al., 2018) analyse how many seeds a deep reinforcement-learning comparison requires. These motivate our seed-level unit of analysis, our internal protocol, and our decision to report an underpowered design explicitly rather than aggregating episodes into an inflated sample size.

Citation verification. 18 of 18 references were checked against publisher-deposited Crossref metadata or the official arXiv record. Source records and corrections are retained in `results/reference_verification.json`. Missing metadata fields are omitted; DOI suffix years are not substituted for publication years.

3. Problem formulation

Consider a metro line operating over a service day divided into decision epochs $t = 1, \ldots, T$. At each epoch the controller observes the state of the line and selects the number of trainsets $a_t \in \{0, \ldots, k_max\}$ to induct into service. Passengers arrive at stations according to a demand process, board subject to train capacity, and are *left behind* when capacity is exhausted.

The operator's objective combines four terms — accumulated waiting, unmet demand, energy and operating cost — through the scalarised reward

```
r_t = −( α·W_t + β·U_t + γ·E_t + δ·C_t )
```

where `W_t` is the queue integral in passenger-minutes, `U_t` the count of distinct left-behind passengers, `E_t` energy and `C_t` operating cost, each normalised against reference scales fitted from the fixed-timetable baseline.

The **primary endpoint** is the `left_behind_fraction`: distinct left-behind passengers as a fraction of arrivals over the service day. **Lower is better.** The primary scenario is `surge`, the condition under which capacity is most likely to bind.

Feasibility — fleet availability, depot capacity and headway constraints — is enforced by **action masking rather than reward penalties**. A penalty makes constraint violation a tunable trade-off; masking makes it unrepresentable.

4. Data and demand forecasting

4.1 Data provenance and classification

We distinguish four categories throughout, because conflating them is the most common way a simulation study overstates its external validity:

Category	In this study
Real data	Metro station-level ridership records, with dataset manifests and licence files recorded
Derived data	Supervised forecasting features and targets constructed from the real records
Synthetic data	The origin–destination distribution (assumption A03); passenger balking and patience b
Simulated data	All operational outcomes — queues, boardings, left-behind counts, energy and cost — p

No operational outcome reported in this paper is a real-world measurement.

4.2 Splits

Splits are chronological with a **7-day embargo** between them: training 2024-01-01 to 2025-10-20, validation 2025-10-28 to 2026-03-23, test 2026-03-31 to 2026-08-13. The embargo exists because adjacent hours are strongly autocorrelated; without it a validation row can be nearly a copy of a training row and the split stops measuring generalisation.

Row counts: 1,338,779 raw station-hours, 5,749 excluded by quality screens, and 1,126,584 / 254,016 / 235,008 supervised training / validation / test examples (Table 1).

Regime screen. A protocol-specified screen sought a run of at least 365 days with year-on-year growth within ±10%. No such run exists in these data; the longest qualifying run was 278 days, so the configured start date was used. This is recorded as a finding rather than applied as a silent default, and all conclusions inherit whatever regime instability remains.

4.3 Forecasting

A quantile gradient-boosted model produces median and quantile forecasts of station-level demand (configuration hash `a976a2d6a212b935`). The observed quantile crossing rate is 0.084767 (Table 2). The historical persistence report records agreement of both median and dispersion against its reference fit. The original binaries and reference predictions are unavailable, so that claim cannot be independently repeated here. The new replication uses a separately validated refit and does not claim identical prediction vectors.

Online inputs and calibration limits. The imported environment appends completed hours of simulated passenger boardings to the forecast history. Boardings depend on dispatch and available capacity, whereas the offline training target is station ridership. Thus past actions can affect subsequent forecasts, and served boardings can be censored relative to arrivals. No future observation is read, but offline interval coverage does not establish online coverage under this feedback. The arm comparison concerns the resulting closed-loop controllers, rather than three policies receiving an identical exogenous forecast path.

5. Metro simulation environment

The environment couples a demand process, rolling-stock and headway constraints, capacity-limited boarding with queue dynamics, and the multi-objective reward of Section 3. An episode is one service day of decision epochs; state transitions advance train positions, board and alight passengers, accumulate queues and record failures.

Validation status across 14 quantities: 4 validated, 3 partially validated, 7 not validatable, 0 mismatch (Table 3).

We describe this as a **simulation environment**, not a validated digital twin. The quantities validated against published operating statistics — running time, round trips per day, minimum headway, weekday boardings — constrain the *supply* side. Of the quantities that could not be validated, four are direct inputs to the crowding side that the primary endpoint measures: Train capacity / peak occupancy, Station dwell-time distribution, Passenger waiting behaviour (balking, patience), Origin-destination distribution.

Consequence, stated once and applied throughout: the *ordering* of methods within the environment is internally valid, because all methods face identical conditions under common random numbers. The *magnitudes* are simulator-internal and must not be read as predictions about a physical metro system.

6. Uncertainty representation and the placebo control

Forecast uncertainty in this study means the **dispersion of the predictive distribution** of near-term demand, summarised as

```
σ = (q90 − q10) / (2 × 1.2816)
```

the normal-equivalent standard deviation implied by the 10th and 90th forecast quantiles. The internal protocol fixes this definition across arms. Its historical registration chronology cannot be independently verified. The online forecast can nevertheless depend on earlier policy actions through its observation history.

Every arm carries the same two-value uncertainty channel and the same parameter count. The arms differ only in that channel's content:

Arm	Channel content
`ppo_point`	Identically zero
`ppo_placebo`	Independent draws with replacement from a rolling pool of up to 50,000 current and previo
`ppo_uq`	The calibrated forecast dispersion σ

What the placebo controls. All arms have identical input width. The placebo resamples values from an empirical pool; it does not preserve a finite sample's marginal distribution exactly. Pool composition changes over time and includes current values, so some demand-regime information can remain. Serial dependence and station-specific associations also differ from the calibrated channel. The pool persists across training episodes but a fresh evaluation environment starts a new pool for each cell. Thus this comparison tests channel constructions, without isolating useful information from every distributional or optimisation effect. No exact permutation or proven information-free control is claimed.

7. Reinforcement-learning framework

- **State.** A 123-value summary comprising station queues, rolling-stock availability, clock features, forecast median and the two-value uncertainty channel. The availability ablation removes the availability block, leaving 121 values.
- **Action.** The number of trainsets to induct in the current epoch, $a_t \in \{0, \ldots, \text{k_max}\}$.
- **Reward.** As in Section 3.
- **Constraints.** Enforced by action masking.
- **Algorithm.** Maskable PPO (Schulman et al., 2017), one frozen hyperparameter configuration applied unchanged to every arm.

No hyperparameter search was conducted. A single documented configuration is applied to all arms. This is a deliberate symmetry constraint: a search conducted on one arm risks selecting a configuration that suits that arm's input, which would be an asymmetry in the comparison. The cost is that all arms may be jointly suboptimal, which bounds any claim about *absolute* policy quality. Section 16 treats this as the study's principal limitation.

Reproducibility of training. Each worker is pinned to a single compute thread, because floating-point reduction order varies with thread count and would otherwise make runs irreproducible. Training days are drawn from a seed pool disjoint from every evaluation, scaler-fitting, tuning and calibration pool (Section 9).

Full PPO configuration values are reproduced in Supplementary Material S2 from the stored policy metadata.

8. Baseline methods

Baseline	Description	Inform
Fixed timetable	A published-style timetable executed without adaptation	Clock o
Threshold heuristic	Induct when queue or forecast exceeds a tuned threshold	Queue,
MILP (static)	Mixed-integer program solved once per day against a fluid-queue surrogate	Foreca
MILP (rolling)	The same formulation re-solved on a rolling horizon	Foreca

A baseline fairness audit of 19 checks passes 19 with 0 blocking failures. Its sharpest check steps three methods through an identical action sequence and asserts identical arrivals and queues, which would detect a common-random-number desynchronisation between methods.

Two aspects of baseline preparation are disclosed because they bear on fairness. The heuristic was **retuned** after its optimum was found at the edge of its search grid; a boundary optimum is not an optimum, so the grid was extended under a recorded amendment until the selection was interior, and every affected result was recomputed rather than mixed with the corrected baseline. The MILP surrogate was **corrected four times** — a standing-queue floor, removal of an unjustified monotonicity clamp, a dispatch-capacity envelope, and earlier service-rate and waiting-coefficient fixes. Each correction strengthened a competitor to the proposed method. A fifth was declined: the fluid-queue surrogate structurally cannot represent departure spacing, and continuing would have meant authoring a baseline rather than calibrating one.

The resulting effort asymmetry — four baseline revisions against one untuned, frozen learned configuration — runs *against* the proposed method, and readers are entitled to weigh it in both directions.

9. Experimental design

- **Unit of analysis:** training seed (protocol section 8.2); episode-level n is never reported as the sample size
- **Registered sample size:** 10 training seeds × 30 evaluation seeds × 4 scenarios
- **Registered episodes:** 6,480, reducing to 216 seed-level observations
- **Stored evaluation cells:** 7,610
- **Main-study learned-policy identities in stored cells:** 50; available original binaries: 0. Pilot observations are excluded.
- **Registered experiments:** 17, in a frozen registry

Evidence classes. The study separates four, and the separation is structural rather than editorial — each stored cell carries a `variant` field that the analysis filters on:

Class	Role
Pilot	Four seeds used only to estimate the between-seed standard deviation for the power gate
Confirmatory	Exactly two registered tests, Holm-corrected at family-wise $\alpha = 0.05$
Exploratory	Hypothesis-generating; Benjamini–Hochberg at $q = 0.10$; supports no claim in the abstrac
Follow-up	A registered training-budget study, **incomplete**; contributes no evidence (Section 16)

Common random numbers. The function constructing an environment's random streams takes the scenario and evaluation seed and **no method argument**, so two methods evaluated at the same seed cannot receive different demand realisations. This is a property of the implementation, not a convention a later edit could silently break.

Seed pools, all mutually disjoint so that nothing is fitted on a day the comparison is later scored on: evaluation [1, 30]; training [101, 110]; pilot training [901, 904]; pilot evaluation [41, 50]; scaler fitting [1001, 1020]; heuristic tuning [2001, 2006]; surrogate calibration [3001, 3004].

Internal registration record. The repository reports that the registry was frozen after the power gate fixed the sample size and before the first confirmatory run. The sweep and ablation stages refuse to execute against an unfrozen registry, making the freeze a precondition of the code rather than a promise in a document.

Sample size and power. A four-seed pilot on disjoint seeds estimated the between-seed standard deviation of the paired difference at 0.005298. Detecting the registered smallest effect size of interest (SESOI = 0.002) at 80% power would require n = 69, beyond the available compute. The registered design proceeded at n = 10 with the underpowering stated in advance.

The pilot variance estimate proved too small, and we report the correction rather than the planning figure. The realised standard deviation of the confirmatory paired differences is 0.011269, **2.13× the pilot estimate**. Four seeds is a thin basis for a variance estimate. Consequently the registered minimum detectable effect of 0.005909 is optimistic, and we do not use it to support the conclusion. **No post-hoc power argument is made**: once a test has rejected, observed power is a deterministic function of the p-value and adds nothing. The study remains underpowered for its own SESOI, so every null reported here is inconclusive rather than evidence of equivalence.

10. Statistical analysis

The repository describes this plan as fixed before the original runs. That chronology is not independently established by this checkout. Here 'confirmatory' names the original protocol's two-test family, not an independently verified registration status.

- **Unit.** Episode outcomes are averaged to one observation per training seed before any test. The sample size is the number of seeds, never the number of episodes; reporting the latter would inflate precision by more than an order of magnitude.
- **Confirmatory family.** Exactly two tests, Holm-corrected at family-wise $\alpha = 0.05$.
- **Exploratory family.** Benjamini–Hochberg at $q = 0.10$.
- **Test selection.** Paired *t*-test where Shapiro–Wilk on the differences does not reject at 0.05 (here $p = 0.6026$); Wilcoxon signed-rank otherwise. Both are reported for the primary comparison.
- **Effect estimation.** Hodges–Lehmann estimator with bias-corrected and accelerated (BCa) bootstrap confidence intervals.
- **Equivalence.** Two one-sided tests against the registered SESOI, run only when the primary test fails to reject — which distinguishes *no evidence of benefit* from *evidence of no meaningful benefit*. The primary test rejected, so no equivalence test applies here.

11. Results

11.1 Method ranking

We report the full ranking first, because it constrains how everything else should be read (Table 6, Figure 9).

Method	normal	surge	disruption	combined
heuristic_threshold	0.00491	0.01184	0.01005	0.01680
milp_static	0.00491	0.01187	0.01005	0.01705
ppo_point	0.01119	0.01794	0.01635	0.02324
ppo_placebo	0.01657	0.02316	0.02019	0.02684
fixed_timetable	0.01879	0.02550	0.02207	0.02864
ppo_uq_noavail	0.02204	0.02905	0.02319	0.03017
milp_rolling	0.02384	0.02932	0.02690	0.03205
ppo_uq *(proposed)*	0.02429	0.03168	0.02563	0.03219
ppo_uq_single_obj	0.13543	0.14694	0.13658	0.14651

Method	Total reward (surge)	Inductions (surge)
heuristic_threshold	-847.4	216.0
milp_static	-857.3	228.0
ppo_point	-956.0	229.0

Method	Total reward (surge)	Inductions (surge)
ppo_placebo	-1035.4	230.9
fixed_timetable	-1042.3	203.9
ppo_uq_noavail	-1123.1	209.9
milp_rolling	-1135.6	228.3
ppo_uq	-1157.0	204.3
ppo_uq_single_obj	-6334.8	195.2

No learned arm outperforms the tuned threshold heuristic or the static mixed-integer formulation in any scenario, on either the primary endpoint or the training objective. The proposed ppo_uq is also worse than the fixed timetable — worse, that is, than not adapting at all.

Baselines are deterministic and carry no training seed, so under the registered seed-level unit of analysis each contributes a single observation and **no inferential test between a learned method and a baseline is defined**. These are differences of means, not hypothesis tests, and are labelled as such throughout. We report them prominently nevertheless: the comparison is unfavourable to the proposed method, and omitting it would misrepresent what the study found.

11.2 Primary comparison (EXP-UQ-001)

ppo_uq versus ppo_point on left_behind_fraction under surge, n = 10 training seeds (Table 5, Figure 4).

Quantity	Value
Mean, ppo_uq	0.031681
Mean, ppo_point	0.017939
Mean paired difference	+0.013742
Hodges–Lehmann estimate	0.013588
95% BCa confidence interval	[0.006570, 0.018523]
Test	paired t-test
Unadjusted p	0.00387057
Holm-adjusted p	0.007741
SESOI (registered)	0.002
Relative degradation	76.6%

Because a **higher** left-behind fraction is worse, a positive difference means the uncertainty-aware arm performed **worse**. The confidence interval excludes zero and lies entirely above the SESOI of 0.002.

This is not a null result. The registered test rejected the null hypothesis in the direction opposite to the one predicted.

11.3 Placebo control (EXP-UQ-002)

ppo_uq versus ppo_placebo: +0.008525, 95% BCa CI [0.002813, 0.014461], Holm-adjusted p = 0.011707 (Table 5, Figure 5).

The mean ordering across the three arms is ppo_point 0.01794 < ppo_placebo 0.02316 < ppo_uq 0.03168. The specified family compares UQ against each comparator; it does not

test placebo against point. Input width is identical, but the bootstrap control does not establish exact distributional matching or complete removal of information. The result supports a difference between these channel constructions, without identifying the learner's internal mechanism.

11.4 Hypothesis verdict

H1 = NOT SUPPORTED.

12. Mechanism analysis

The documented design changes uncertainty-channel content while holding the specified learner and random-number scheme fixed. Episode-level induction totals associate the informative channel with fewer dispatches. The historical per-epoch records and model binaries are absent from this checkout, so their analyses cannot currently be rerun. This is a behavioural association across separately trained policies, not a test of the within-policy causal effect of changing uncertainty at a fixed point forecast. Nothing was re-simulated for this revision.

12.1 Informative uncertainty suppresses dispatch; placebo uncertainty does not

Scenario	Calibrated σ vs zero	p	Placebo σ vs zero	p
normal	-26.3 inductions	7.4e-06	+1.8	0.23
surge	-24.7 inductions	9.6e-06	+1.9	0.20
disruption	-14.9 inductions	3.6e-04	+1.9	0.30
combined	-14.1 inductions	4.2e-04	+1.9	0.27

Under surge this is approximately a **11% reduction in dispatch volume** relative to the point-forecast arm (Figure 6). The placebo arm does not move. The evidence is therefore consistent with the controller responding to genuine forecast dispersion, and responding by withholding service.

12.2 Dispatch timing, at matched volume

`ppo_uq` and the fixed timetable release almost the same number of trains, which makes them the one pair in the study comparable at approximately matched volume:

Quantity	`ppo_uq`	`fixed_timetable`
inductions	204.3	203.9
idle epochs	84.4	54.1
max queue	1195.1	980.6

At comparable volume the uncertainty-aware arm leaves markedly more decision epochs with no dispatch and reaches a larger peak queue than a non-adaptive schedule. This suggests both restraint and clustering are present. Across methods the relationship between dispatch volume and service is not monotone — the tuned heuristic achieves the best service in the study while dispatching fewer trains than `ppo_point` — so timing matters alongside volume.

12.3 Nonlinear service-loss association and its limits

The historical Figure 7 bins arrivals and left-behind counts across operating epochs. This is an association across different states and decisions. Dispatch volume, train capacity, queue history and time of day are not held fixed; the curve therefore does not establish a structural convex loss function or an optimal dispatch response. Its underlying epoch records are unavailable in this checkout.

For an independently justified fixed convex loss ℓ, Jensen's inequality gives $E[\ell(D)] \geq \ell(E[D])$. A mean-preserving spread ordered in the convex order increases expected convex loss; a change in variance alone does not establish that ordering. The newsvendor analogy (Arrow et al., 1951) motivates a future decision-theoretic analysis, but it neither derives the optimum under our multi-objective reward nor proves that this policy should dispatch upward when dispersion increases.

Interpretation, stated with its limits. We separate what was observed from what it suggests:

Observed	Informative σ is associated with fewer dispatches and worse service; placebo σ is
Mechanism evidence	The placebo comparison distinguishes these channel constructions at identical in
Interpretation	The evidence is *consistent with* the controller having learned a association betwe
Limitation	We do not observe the network's internal computation. The registered epoch-leve
Future hypothesis	If this is an optimisation failure rather than a property of the information, a large

We deliberately do not claim the controller *learned the wrong sign* as a mathematical statement. The available evidence supports a behavioural description and a plausible reading of it, not a proof about the learned value function.

13. Robustness and sensitivity

Check	Result
Scenarios reproducing the direction	4/4
Training seeds favouring `ppo_point`	9/10
Distribution-free cross-check (Wilcoxon signed-rank)	p = 0.005859
Worst leave-one-seed-out replicate	p = 0.01021 (seed 107), Holm-adjusted 0.02043
Sensitivity conditions where `ppo_uq` is worse	12/12

The conclusion does not depend on any single training seed, on the choice of parametric test, or on one scenario (Table 9, Figure 8).

Origin–destination sensitivity. The OD structure is synthetic and drives the primary endpoint, so the comparison was re-scored under all three OD variants. The registered direction holds in every one, against both comparators (Table 7, Figure 10). Individual conditions mostly do not clear the Benjamini–Hochberg threshold, which is expected — each uses five training seeds and ten evaluation seeds against ten and thirty in the main sweep — so the evidence is the consistency of sign rather than any single test. These conditions re-score the *same* policies and are therefore strongly dependent; consistency across them is weaker evidence than independent replication would be.

Reward-weight sensitivity is vacuous on the primary endpoint, and we report that rather than the analysis it appears to be. Re-scoring a frozen policy under a different

reward weighting changes no action, because the policy does not observe the reward at evaluation time. The trajectory — and with it `left_behind_fraction` and `inductions` — is identical across all three weightings, verified per seed; only `total_reward` differs. Measured there, `ppo_point` wins under every weighting, by a margin that grows as the weighting favours service, which is the ordering the under-dispatching mechanism predicts. Testing weight sensitivity on physical outcomes would require *retraining* under each weighting, which this design did not do.

Interpretability. Feature attributions for one policy at one training seed are reported in Supplementary Material S9 (shap.PermutationExplainer (model-agnostic)). They describe model association on a chosen instance set and **do not establish causality**; they support no hypothesis in this paper.

14. Discussion

What was expected. Demand uncertainty is decision-relevant in capacity-constrained service systems, and the registered hypothesis was that supplying it to the controller would reduce the fraction of passengers left behind.

What was observed. The opposite, significantly and consistently. The uncertainty-aware controller left more passengers behind than the point-forecast controller, and more than a past-value bootstrap placebo. It also dispatched substantially fewer trains when given calibrated dispersion.

Why it may have happened. Three readings are consistent with the evidence, and the design separates them only partly.

Reading 1 — the signal is genuinely unhelpful in this setting. The placebo comparison shows that calibrated σ performed worse than a resampled channel at identical input width. However, changed temporal dependence and imperfect distributional matching prevent a unique attribution to information quality.

Reading 2 — the information is usable but the controller did not learn to use it. One frozen configuration at a fixed budget may fail to exploit an input that a longer or better-tuned run would. No learned arm beats a threshold heuristic, which motivates this alternative explanation but does not establish it. The incomplete budget follow-up does not yet distinguish these explanations.

Reading 3 — the scalarisation misdirects the response. Under a weighting where energy and cost carry real weight, dispatch has an immediate certain cost while its benefit is delayed and uncertain. A controller reading high dispersion as elevated risk may protect the certain terms and lose more on the service terms. The observed pattern — lower energy and cost, worse service, worse total reward — is consistent with this.

Distinguishing readings 1 and 2 requires a training-budget experiment. Until it is run, the defensible statement is the narrow one: *in this environment, at this budget, the uncertainty channel did not help and measurably hurt.*

What it means for uncertainty-aware reinforcement learning. It identifies a concrete failure mode — an informative uncertainty input producing a systematically conservative response associated with worse service in the tested environment — and shows that the gap survives a placebo control. It does **not** establish that uncertainty-aware reinforcement learning is generally ineffective, and no result here speaks to risk-sensitive objectives, distributional value functions, or architectures designed to consume uncertainty explicitly.

What it means for evaluation practice. A single-seed, episode-level comparison without a placebo could plausibly have produced a favourable and unreplicable result from the same

experimental material (Henderson et al., 2018; Colas et al., 2018). The control structure, not the algorithm, is what makes this study's conclusion interpretable.

15. Practical implications

Stated conditionally, because the environment's crowding components are not externally validated and the study covers one line.

For metro operators. Adding a calibrated uncertainty feature to a learned dispatch controller should not be assumed to improve service. Evaluation should include a placebo-controlled comparison rather than a before/after improvement, and should stress the controller in **capacity-binding regimes**, where the cost of a conservative decision is nonlinear and where a well-behaved controller in slack conditions may still fail.

For transit planners. In this study a tuned threshold heuristic outperformed every learned controller. Where a simple demand-responsive rule is available, it is a serious operational candidate and an essential evaluation baseline, not a formality.

For reinforcement-learning researchers. The failure mode is specific and testable: an informative uncertainty input was associated with fewer dispatches and worse service. Reporting dispatch-volume behaviour alongside outcome metrics would make such effects visible earlier.

For forecasting researchers. Calibration is necessary but not sufficient. A well-calibrated dispersion estimate can still degrade a downstream controller if the controller's response to it is mis-signed.

For simulation researchers. The quantities that determine an endpoint deserve validation priority. Here the supply-side quantities were validated and the crowding-side quantities — which define the endpoint — largely were not.

No claim of deployment readiness is made. Nothing in this study licenses operational use.

16. Limitations

1. **Underpowered for the registered SESOI.** n = 10 against n = 69 required at the pilot standard deviation, and the realised standard deviation was 2.13× larger still. Every null result here is inconclusive, never evidence of equivalence.
2. **A single frozen training configuration**, which may be undertrained in absolute terms rather than merely relative to its siblings: no learned arm beats the threshold heuristic, and one single-objective training seed diverged by two orders of magnitude. This is the largest threat to the interpretation and is not resolvable by re-analysis.
3. **Training-budget confounding.** The study cannot separate *the uncertainty channel is unhelpful* from *this optimiser could not exploit it at this budget*.
4. **The learned controllers underperform a tuned heuristic**, which bounds the practical contribution to nothing and must be read alongside the primary result.
5. **7 of 14 environment quantities are not externally validated**, including four that feed the primary endpoint directly. Magnitudes are simulator-internal.
6. **Synthetic origin–destination structure**, mitigated but not eliminated by the OD sensitivity analysis.
7. **A single line, operator and network configuration.** No external-validity claim.
8. **Limited external validation** of the environment overall; it is a simulation environment, not a validated digital twin.
9. **The registered follow-up is incomplete.** The training-budget study was stopped at 11 of 80 policies, at a single budget, asymmetrically distributed across arms, with **no evaluation cells**. It supports **no result and none is reported from it**: not even the

lowest budget admits a paired comparison. It appears in this paper only as an open question.
10. **No transfer or generalisation evidence.** Nothing here indicates how the finding would transfer to another line, demand regime, architecture or reward.
11. **The registered mechanism test (epoch-level) is unresolved**, because stored cells do not carry the per-epoch forecast.
12. **Feature attributions are associational**, computed for one policy at one seed.
13. **The bootstrap placebo is imperfect.** It samples a changing empirical pool with replacement, rather than permuting a fixed batch. Exact marginal matching and complete removal of information are not established; differences in temporal dependence can affect learning.

These are collected in Table 10.

17. Future work

The following are **proposed, not completed**. Follow-up planning occurred after the original outcome was known and is exploratory. It cannot retroactively alter the original family or its conclusion. The historical follow-up remains blocked by missing original models. A separate independent replication is specified under `docs/replication_20261015_protocol.md`, using fresh anchors and an explicitly updated source snapshot. See `docs/replication_execution.md` for its execution status. It provides no new result until the complete grid passes its checks.

1. **Registered training-budget sweep.** The single most informative remaining experiment for testing budget dependence of the contrast. It cannot alone establish information quality or optimality. A protocol and a specified analysis exist in the repository; execution is incomplete.
2. **Symmetric hyperparameter selection.** Select one configuration on independent validation data using the *point-forecast* arm only, then apply it unchanged to all arms. Tuning on the arm that is not the hypothesis preserves symmetry while addressing the absolute-undertraining concern.
3. **Uncertainty-scaling probe.** Supply a deliberately mis-scaled dispersion signal. A monotone response would implicate magnitude rather than information content.
4. **Risk-sensitive and robust formulations.** Test controllers designed to consume uncertainty explicitly, rather than supplying uncertainty to a risk-neutral learner.
5. **Alternative reward weightings, with retraining**, to test whether the conservative response persists when unmet demand dominates the objective.
6. **Multi-line and broader demand regimes**, and validation of the crowding-side environment components against operator data.
7. **Operational pilot evaluation** with human dispatchers, which no simulation result can substitute for.

18. Reproducibility and software artefact

Item	Value
Git commit	`e053dea4180cec859f7241c1ce529e0d7235a8ac`
Original-study tag in this checkout	`unavailable in this checkout`
Interpreter	Python 3.11.9 on Windows-10-10.0.26200-SP0
Dependencies	55 pinned packages, 1047 SHA-256 hashes
Evaluation cells	7,610
Trained policy binaries available here	0 in the registered namespace; historical metada
Registered experiments	17
Historical reproducibility report (not current validation)	PASS = 23, FAIL = 0, WARN = 1, SKIPPED = 0

Item	Value
Historical immutability report (tag unavailable here)	PASS = 14, FAIL = 0

Stored-results validation on Linux uses a separate CPU hash lock; it does not reconstruct missing historical inputs or prove experiment reproducibility:

```
uv venv --python 3.11 .venv
uv pip install --python .venv/bin/python --require-hashes --torch-backend cpu -r requirements-l
.venv/bin/python scripts/check_environment.py
.venv/bin/python -m pytest -q
.venv/bin/python scripts/publication_audit.py  # rechecks stored cells; nonzero until blockers
```

Every figure carries a `.source.json` provenance companion naming the artefacts it was drawn from, and this manuscript is rendered by a script that reads those artefacts, so no reported number is typed by hand. Full instructions are in the repository's reproduction guide.

Provenance caveat, stated because a reproducibility claim a reader cannot act on is worse than a stated limitation. Almost all stored evaluation cells were produced while the working tree contained uncommitted changes; their manifests record this as `provisional`. The recorded commit therefore does not fully determine the code that produced them. The current import and uncommitted revision cannot reconstruct that historical working tree. New computation requires its own frozen source/input snapshot.

The available repository contains one initial import commit; commits named in original cell manifests and the original-study tag are absent. The current audit reports missing raw files, frozen simulator inputs, forecasting models and policy binaries explicitly. A historical audit's PASS does not establish availability now. Restoring the exact original artifacts is a prerequisite for the budget follow-up; using a newer GTFS feed would change a held-fixed input. The reported BCa intervals estimate the Hodges–Lehmann paired location, not the mean difference.

Software artefact. The repository includes two interactive applications: a research results dashboard, and an operator-facing decision-support prototype that presents the simulated line state, produces dispatch recommendations from the trained policies and compares strategies. Both are demonstration artefacts served locally; **neither constitutes deployment**, and the prototype does not control railway infrastructure. The prototype reads research artefacts read-only, and its interface labels every displayed quantity with its evidence class so that a simulated demonstration value cannot be read as a research result.

19. Conclusion

We evaluated whether supplying calibrated demand uncertainty to a reinforcement-learning metro dispatch controller improves passenger service, in an internally specified, placebo-controlled experiment on a demand-driven simulation environment.

The registered hypothesis was **not supported**. The uncertainty-aware controller left more passengers behind than an otherwise identical point-forecast controller (+0.013742; 95% BCa CI [0.006570, 0.018523]; Holm-adjusted p = 0.007741), and also performed worse than a placebo carrying dispersion values resampled from a rolling empirical pool.

Rather than improving service, the controller exhibited a systematic conservative dispatch response to genuine forecast uncertainty — approximately 11% fewer dispatches under surge — which the placebo arm did not detectably differ from zero. The episode totals do not identify an optimal dispatch direction or a causal within-policy response to dispersion. Combined with the finding that no learned arm outperformed a tuned threshold heuristic, the evidence

points toward a policy-learning limitation rather than toward uncertainty information being valueless.

Controlled negative evidence is informative when the design distinguishes competing explanations. Here a placebo arm and a seed-level unit of analysis were sufficient to overturn an intuition that a less controlled comparison would likely have confirmed. The result characterises the tested controller, configuration, training budget, environment and scenarios; it identifies a concrete failure mode and motivates controlled investigation of uncertainty utilisation, training budget and risk-sensitive policy design.

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Supplementary material: `docs/supplementary_material.md`.

Figures

Figure 3 was corrected from stored forecast metrics in this revision; Figures 1 and 5 were regenerated with corrected placebo labels. Other displayed plots are historical files. Their presence does not mean the source computations were rerun; Figure 7 is withheld as explained below.

Figure 1. research framework

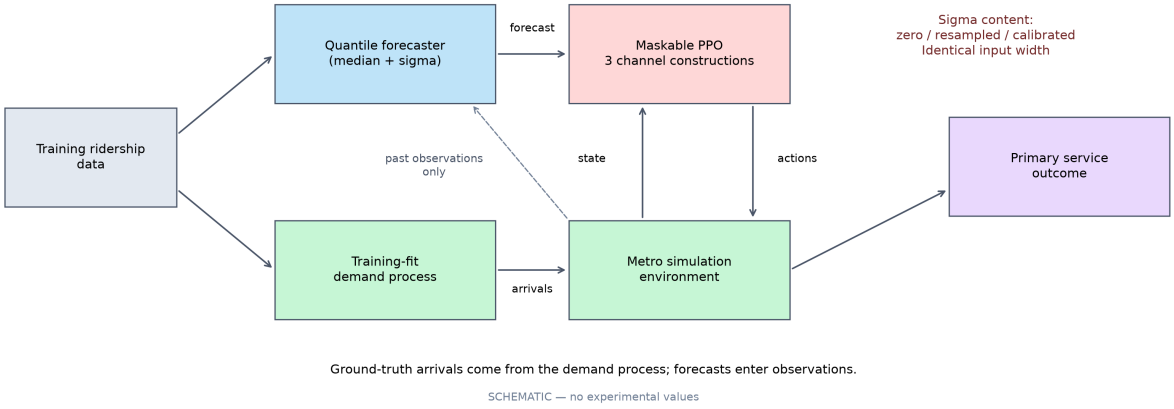
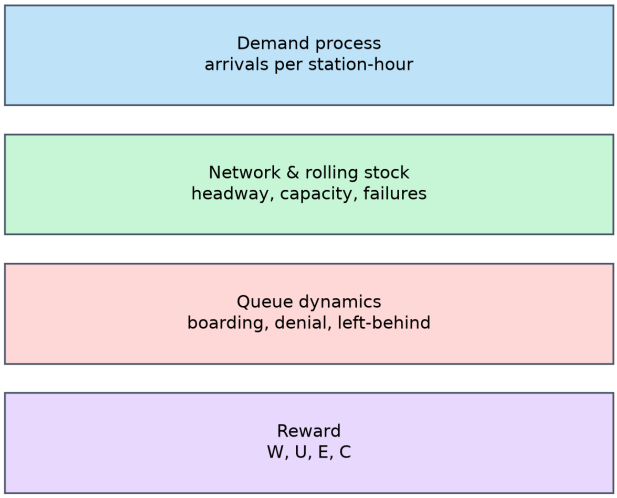


Figure 2. twin architecture



External validation status (14 quantities)

4	validated
3	partially validated
7	not validatable
0	mismatch

The four not-validatable quantities that feed the primary endpoint are capacity/occupancy, dwell, balking, OD.

Figure 3. forecasting

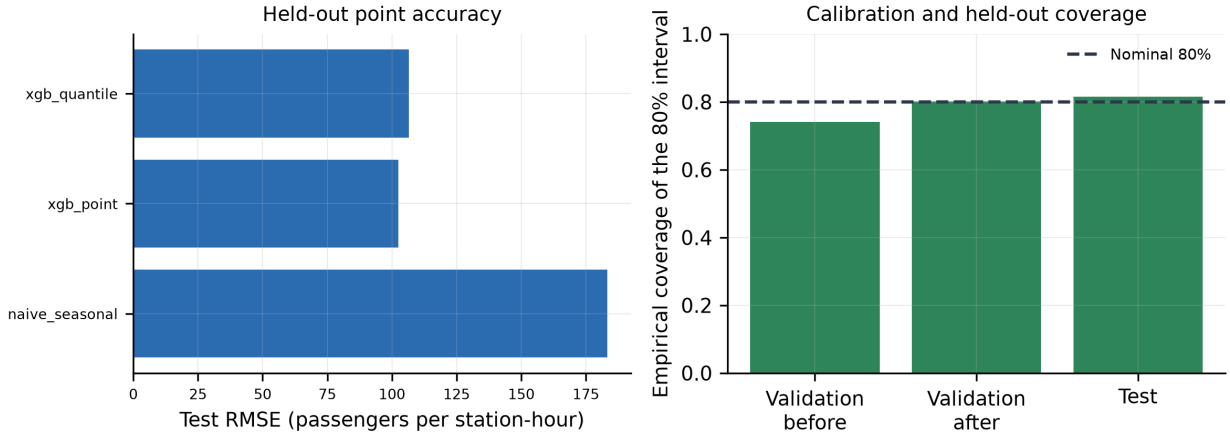


Figure 4. primary uq vs point

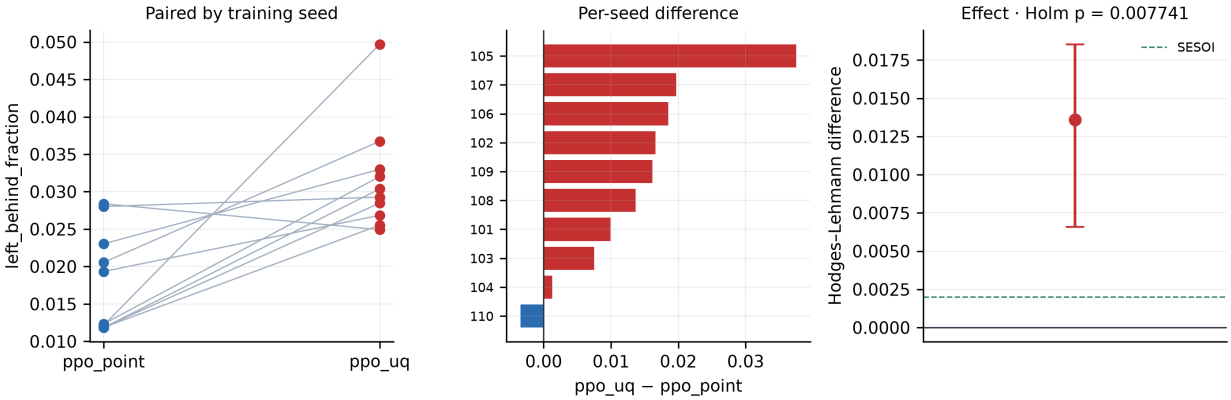


Figure 5. placebo

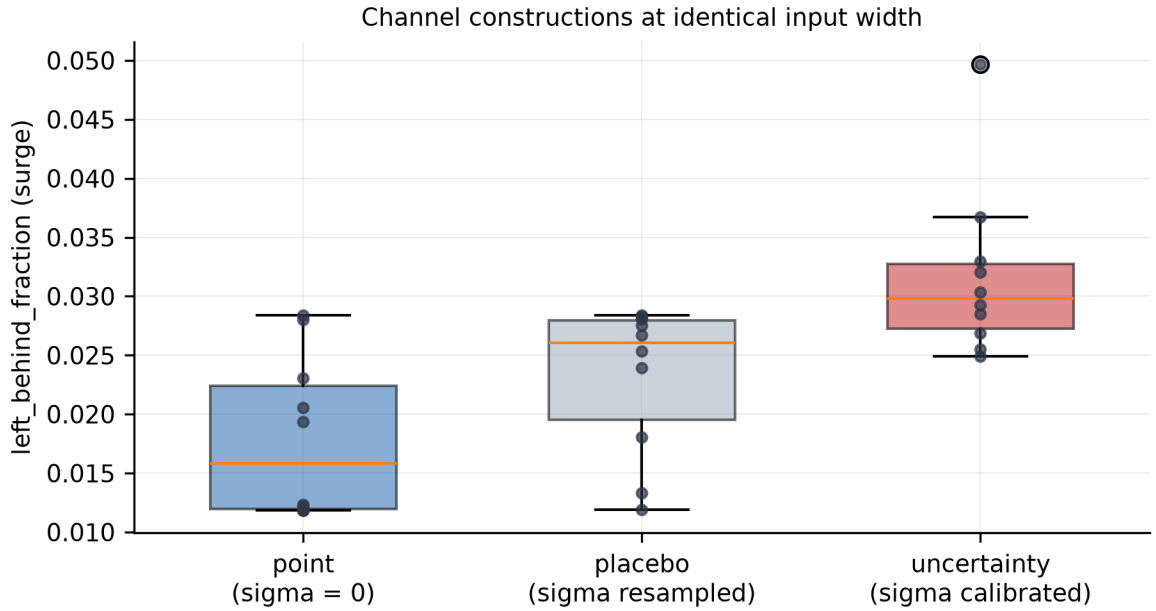


Figure 6. mechanism

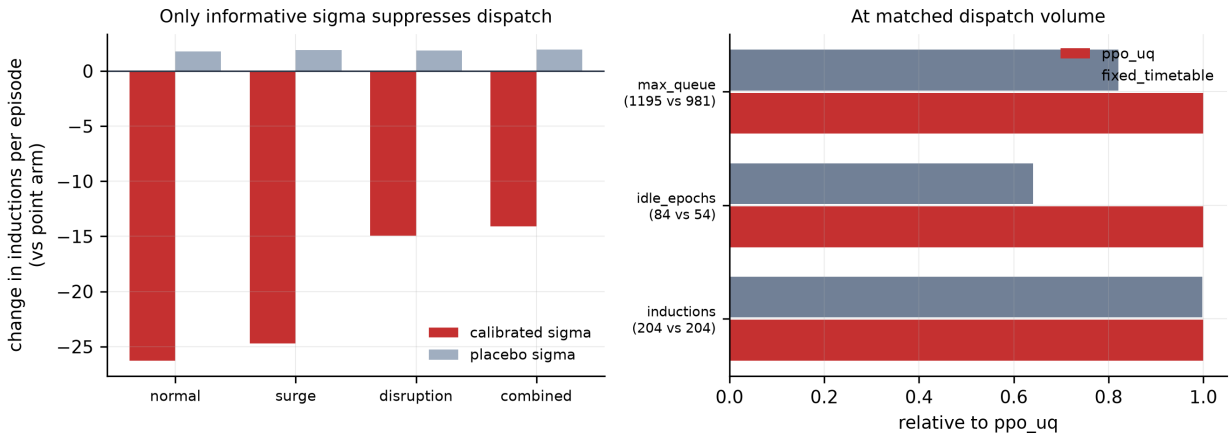


Figure 7. Historical arrival/loss curve — withheld from this revision

The archived plot's epoch-level source data are unavailable, and its convexity interpretation is not identified by binned observational summaries. The original file is retained in results/paper_figures; it is not presented here as evidence of a structural loss function.

Figure 8. scenario robustness

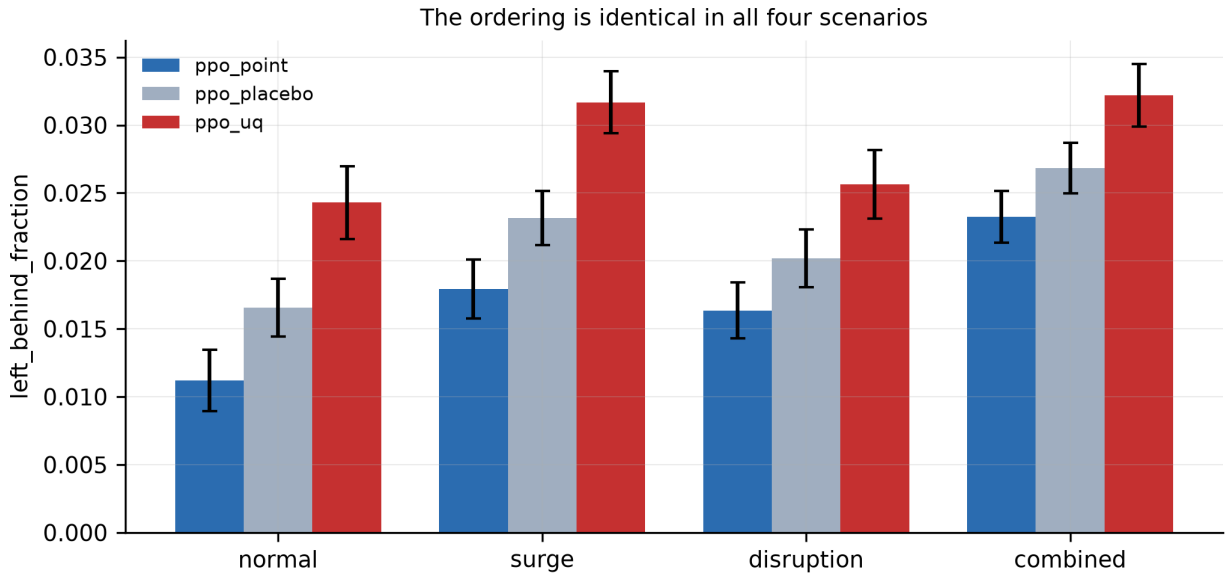


Figure 9. baselines

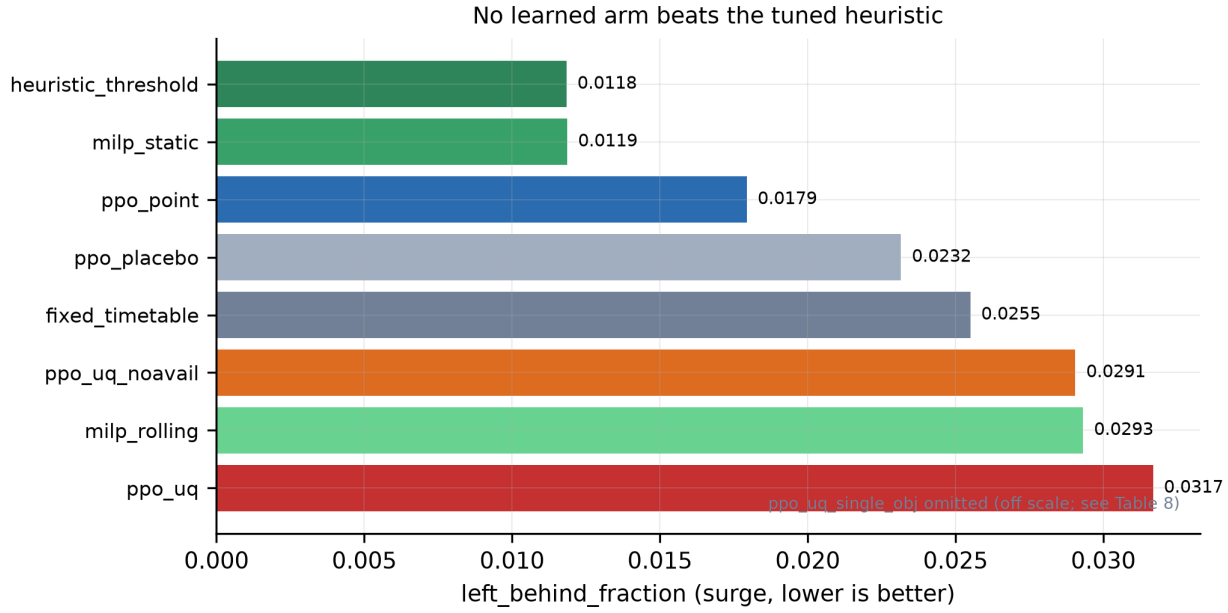


Figure 10. sensitivity

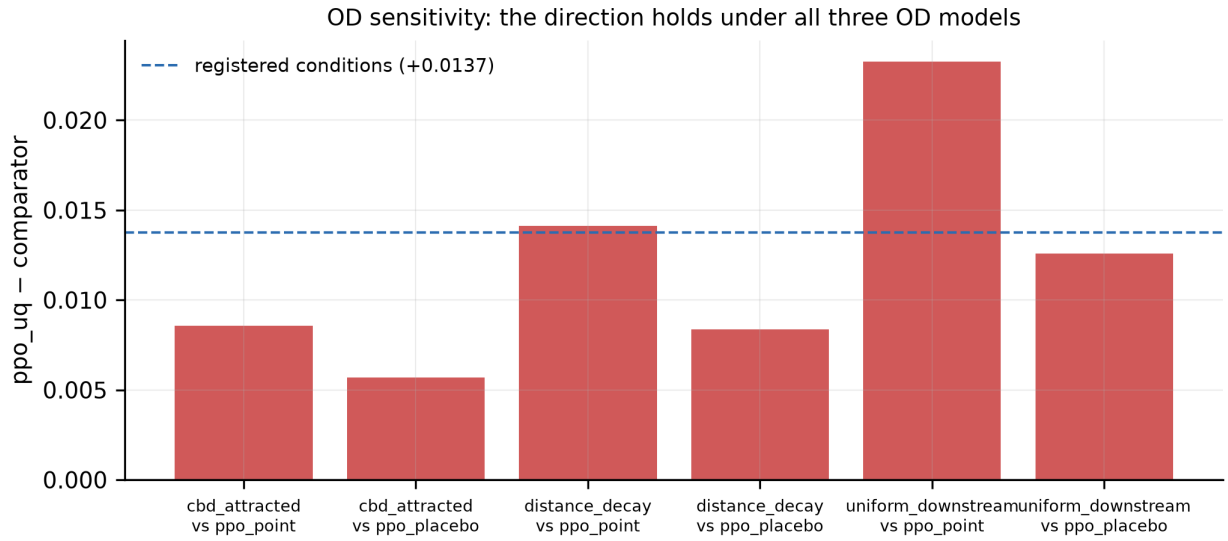


Figure 11. shap

